

PROJECT 6

Drawing machine

It's time to try a more complex project. This program, the drawing machine, turns a string of simple instructions into turtle commands to draw different shapes. The skills used in planning this program are essential for any coder.

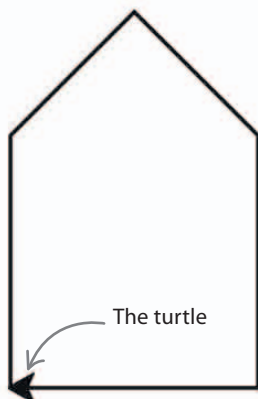
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Choose a test shape

To write a program that can draw any shape, it's useful to choose a shape to start with. Use this house shape as an example to test the program at each stage. By the end of the project it will be possible to draw this house with far less code—by using a single string containing several short drawing commands (for example, "F100").



▷ Turtle draws a house

The arrow shows the final direction and position of the turtle. Starting at the bottom left, it has moved clockwise around the house.

```
from turtle import *
reset()
left(90)
forward(100)
right(45)
forward(70)
right(90)
forward(70)
right(45)
forward(100)
right(90)
forward(100)
```

Loads all the commands that control the turtle

Resets the turtle's position and puts the pen down ready to draw

Moves the turtle forward by 70

Makes the turtle turn 90 degrees to the right

△ Program to draw a house

This code tells the turtle to draw a house. It requires lots of lines of code for what is actually quite a simple program.

Three parts of the program

The drawing machine will be a large program. To help with the planning, it can be broken down into three parts, each one related to a different task.

Function 1

△ Turtle controller

This function takes a simple command from the user and turns it into a turtle command. The user command will come as a single letter and a number.

Function 2

△ String artist

In this program, the user enters a string of instructions. This function splits the string into smaller units, which are then fed to the Turtle controller.

Main program

△ User interface

The String artist needs to get its input from somewhere. The User interface allows the user to type in a string of commands for the String artist to work on.